

Foundation (Jalapeno)

- Q1.** What is the purpose of a two-way frequency table?
(A) To compare two variables
(B) To calculate mean and median
(C) To display a single variable
(D) To calculate standard deviation
(E) To create a bar graph
- Q2.** If a and b are variables in a two-way table, what does $P(acapb)$ represent?
(A) Probability of a or b
(B) Probability of a and b
(C) Probability of a given b
(D) Probability of b given a
(E) None of the above
- Q3.** What is the formula to calculate relative frequency?
(A) $\frac{\text{frequency}}{\text{total observations}}$
(B) $\frac{\text{total observations}}{\text{frequency}}$
(C) $\text{frequency} \cdot \text{total observations}$
(D) $\frac{\text{frequency}}{\text{mean}}$
(E) $\text{total observations} - \text{frequency}$
- Q4.** Which of the following is NOT an association between variables?
(A) Positive association
(B) Negative association
(C) No association
(D) Inverse association
(E) All of the above are associations
- Q5.** If the relative frequency of a and b occurring together is 0.6, what is $P(acapb)$?
(A) 0.4
(B) 0.6
(C) 0.8
(D) 0.2
(E) 1.0
- Q6.** In a two-way table, what does a value of 1 in a cell represent?
(A) One observation
(B) Zero observations
(C) Two observations
(D) No observations
(E) Total observations
- Q7.** What type of graph is best used to display a two-way table?
(A) Bar graph
(B) Histogram
(C) Scatter plot
(D) Box plot
(E) Heat map
- Q8.** If $P(a) = 0.4$ and $P(b) = 0.6$, what is $P(acupb)$ if a and b are mutually exclusive?
(A) $0.4 + 0.6$
(B) $0.4 \cdot 0.6$
(C) $0.4 - 0.6$
(D) 1.0
(E) 0.2

Open Ended Questions (FRQ)

- Q9.** Suppose we have the following two-way table:

	b_1	b_2	Total
a_1	10	20	30
a_2	15	25	40
Total	25	45	70

 Calculate the relative frequency of a_1 and b_1 .
- Q10.** A survey of 100 students found the following results:

	Like	Dislike	Total
Boys	30	20	50
Girls	25	25	50
Total	55	45	100

 Analyze the association between the variables 'gender' and 'preference'.

Chef's Tips

- Tip 1: Make sure students understand the concept of two-way frequency tables before moving on to relative frequencies and associations.
- Tip 2: Use real-life examples to illustrate the use of two-way tables.
- Tip 3: Remind students that $P(acapb) = P(a) \cdot P(b)$ only if a and b are independent.